

# Review article: Numerical Methods for Engineering Quantum Computers

Shu Liu<sup>a</sup>, Antigoni Georgiadou<sup>b</sup>, Spancer Dong<sup>b</sup>, Eric Kubischta<sup>a</sup>, Daniel Melton<sup>a</sup>, Stefanie Guenther<sup>c</sup>, Shi Jin<sup>d</sup>, Lin Lin<sup>e</sup>, Philip Lotshaw<sup>b</sup>, Anders Petersson<sup>c</sup> and Mark Sussman<sup>a,\*</sup>

<sup>a</sup>Department of Mathematics, Florida State University, Tallahassee, 32306, Florida, USA

<sup>b</sup>Oak Ridge National Laboratory, Oak Ridge, 37830, Tennessee, USA

<sup>c</sup>Lawrence Livermore National Laboratory, Livermore, 94550, California, USA

<sup>d</sup>Shanghai Jiao Tong University, Shanghai, 200240, China

<sup>e</sup>Department of Mathematics, University of California, Berkeley, 94720, California, USA

## ARTICLE INFO

### Keywords:

quadrupole exciton  
polariton  
WGM  
BEC

## ABSTRACT

In the article by Herrmann et al, “Quantum utility – definition and assessment of a practical quantum advantage,” the quantum computer “Application Readiness Level” (ARL) is defined: (ARL1) “concept with unknown potential,” (ARL2) “beneficial in small idealized systems,” (ARL3) “utility indicated by theory or resource estimations,” (ARL4) “simulated utility demonstration” and (ARL5) “utility demonstration on quantum hardware.” Herrmann et al posit that ARL3 has been reached via demonstrations of the Variational Quantum Eigenvalue (VQE) algorithm. In this review article, the latest activities from the computational “digital twin” perspective are reported on in which scientists are working hard to bring practitioners out of the Noisy Intermediate Scale Quantum (NISQ) era into an era of “Quantum supremacy.” In other words, this review article discusses digital twin activities that help to advance from ARL3 to ARL4, or even ARL5. The three main categories in this review article are (1) Optimization, Control, Inverse problems, and exploratory simulation algorithms for Schrodinger’s equation, (2) Quantum computer Error correction, and (3) Quantum algorithms for benchmarking quantum computers (to include emulator algorithms like Qiskit).

## 1. Introduction

The motivation for this review article is the fact that Moore’s law is no longer applicable [94]. As quoted from Shalf [115], “Moore’s Law [1] is a techno-economic model that has enabled the IT industry to double the performance and functionality of digital electronics roughly every 2 years within a fixed cost, power and area. This expectation has led to a relatively stable ecosystem (e.g. electronic design automation tools, compilers, simulators and emulators) built around general-purpose processor technologies, such as the x86, ARM and Power instruction set architectures. However, within a decade, the technological underpinnings for the process that Gordon Moore described will come to an end, as lithography gets down to atomic scale. At that point, it will be feasible to create lithographically produced devices with dimensions nearing atomic scale, where a dozen or fewer silicon atoms are present across critical device features, and will therefore represent a practical limit for implementing logic gates for digital computing [2]. Indeed, the ITRS (International Technology Roadmap for Semiconductors), which has tracked the historical improvements over the past 30 years, has projected no improvements beyond 2021, as shown in figure 1, and subsequently disbanded, having no further purpose. The classical technological driver that has underpinned Moore’s Law for the past 50 years is failing [3] and is anticipated to flatten by 2025, as shown in figure 2. Evolving technology in the absence of Moore’s

\*Corresponding author

 sliu11@fsu.edu (S. Liu); georgiadoua@ornl.gov (A. Georgiadou); dongx@ornl.org (S. Dong); ekubischta@fsu.edu (E. Kubischta); dnm21b@fsu.edu (D. Melton); guenther5@llnl.gov (S. Guenther); shijin-m@sjtu.edu.cn (S. Jin); linlin@math.berkeley.edu (L. Lin); lotshawpc@ornl.gov (P. Lotshaw); petersson1@llnl.gov (A. Petersson); msussman@fsu.edu (M. Sussman)

ORCID(s): <https://orcid.org/0000-0003-1009-7726> (S. Liu); <https://orcid.org/0000-0001-5451-0290> (E. Kubischta); <https://orcid.org/0000-0002-6116-3722> (S. Guenther); <https://orcid.org/0009-0006-2337-201X> (S. Jin); <https://orcid.org/0000-0001-6860-9566> (L. Lin); <https://orcid.org/0000-0002-7594-2735> (P. Lotshaw); <https://orcid.org/0000-0003-0522-7872> (A. Petersson); <https://orcid.org/0000-0001-8238-4295> (M. Sussman)

Law will require an investment now in computer architecture and the basic sciences (including materials science), to study candidate replacement materials and alternative device physics to foster continued technology scaling.”

References one through three in the above quote correspond to the following references respectively: [94], [89], [91].

As further motivation for this review article, in Figure 3 of the article by Shalf [115], a roadmap is provided for possible paths forward when the density of circuits on classical computers exceeds a critical value. There are three categories: (a) roadmap for the next ten years, (b) 20 years, and (c) “Decades beyond exascale,” “New Models of Computation.” For category (c), Shalf [115] refers to the following prospective technology: (i) approximate computing, (ii) adiabatic reversible, (iii) Analog, (iv) Neuromorphic, and (v) quantum [36, 37].

Of the “new models of computation” listed in Shalf’s article [115], this review article will address the “Numerical Methods for Engineering Quantum Computers” aspects associated with the emerging “quantum computing” paradigm. As outlined by Gamble [40], the development of reliable quantum computers will have a transformative effect on computer technology. A perfectly working fifty qubit quantum computer can efficiently process  $2^{50}$  pieces of data which is 1024 Terabytes (1 Petabyte)!

That being said, right now we are in the “Noisy Intermediate-Scale Quantum” [21] (NISQ) era. The purpose of this review article is to bring to the forefront the current research being done, from the computational perspective, in order to move beyond the NISQ era.

## 2. Numerical methods for computing solutions to Schrodinger’s equation and Numerical methods for optimization and control in which the objective function corresponds to PDE constrained optimization with respect to Schrodinger’s PDE equation

In 1982, Feit et al [34] developed a spectral method for determining the eigenvalues and eigenfunctions of the time independent Schrodinger equation:

$$i \frac{\partial \psi}{\partial t} = -\frac{1}{2M} \nabla^2 \psi + V(x, y) \psi \quad (1)$$

$$\psi(x, y, t) = \sum_{n,j} A_{nj} u_{nj}(x, y) e^{-i E_n t} \quad (2)$$

$$E_n u_{nj} = -\frac{1}{2M} \nabla^2 u_{nj} + V u_{nj}, \quad (3)$$

where

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}. \quad (4)$$

In 2002, Bao et al [6] developed a time splitting spectral approximation for the time dependent Schrodinger equation in the “semiclassical” regime:

$$i u_t^\epsilon = -\frac{\epsilon}{2} \Delta u^\epsilon + \frac{1}{\epsilon} V(x) u^\epsilon \quad x \in \mathbb{R}^d \quad u^\epsilon(x, 0) = u_0^\epsilon(x) \quad (5)$$

Bao et al report results in one and two space dimensions.

In 2004, Zheng et al [136] developed a finite element method for finding the ground state of the lithium atom. In otherwords, Zheng et al found the smallest eigenvalue and the associated eigenfunction for the time independent Schrodinger equation:

$$E\psi = H\psi \quad (6)$$

where the Hamiltonian operator  $H$  is defined as

$$H = -\frac{1}{2}(\Delta_1 + \Delta_2 + \Delta_3) - \frac{3}{r_1} - \frac{3}{r_2} - \frac{3}{r_3} + \frac{1}{r_{12}} + \frac{1}{r_{13}} + \frac{1}{r_{23}} \quad (7)$$

$$\Delta_j = \frac{\partial^2}{\partial x_j^2} + \frac{\partial^2}{\partial y_j^2} + \frac{\partial^2}{\partial z_j^2} \quad 1 \leq j \leq 3 \quad (8)$$

$$r_{jk} = |\vec{r}_j - \vec{r}_k| \quad r_j = |\vec{r}_j| \quad (9)$$

After converting from cartesian coordinate system to spherical coordinate system, the authors were able to reduce the dimension of the problem from 9 to 6.

In Grivopoulos' 2005 Phd thesis [44], methods were developed for the control of quantum systems. We highlight one such control problem discussed in the thesis. The Lindblad equation [90, 82] for open quantum systems is (see 3.7 in [44]):

$$\dot{\rho} = -i[H, \rho] + \sum_n K_n \rho K_n' - \frac{1}{2} \left\{ \sum_n K_n' K_n, \rho \right\} \quad (10)$$

The Lindblad equation models the interaction of two quantum systems, A and B, with respective Hilbert spaces  $H_A$  and  $H_B$ . The Hilbert space for the composite space is,

$$H_{AB} = H_A \otimes H_B \quad (11)$$

A general state vector on the composite space is,

$$\psi_{AB} = \psi_A \otimes \psi_B. \quad (12)$$

As stated in [44], "For the open system A (interacting with its environment B),  $\rho$  is the state variable analogous to the vector  $\psi$  for an isolated quantum system." Expressions for  $\rho$  are:

$$\rho = \text{tr}_B \psi_{AB} \psi_{AB}' = \sum_{ij} \left( \sum_I (\psi_{AB})_{iI}^* (\psi_{AB})_{jI} \right) e_j e_i' \quad (13)$$

Note that  $\rho$  is self-adjoint, positive semi-definite, and  $\text{tr} \rho = 1$ . As an example of control of open quantum systems, [44], considers a control problem for the laser cooling of atoms and molecules. The Lindblad equation with Hamiltonian control is

$$\dot{\rho} = -i[H_0 + V u(t), \rho] + L(\rho), \quad (14)$$

where

$$L(\rho) = \sum_{ij} K_{ij} \rho K_{ij}^\dagger - \frac{1}{2} \{ K_{ij}^\dagger K_{ij}, \rho \} \quad (15)$$

[44], for the  $3 \times 3$  case, determines the control,  $u(t)$ , in (14), that maximizes  $\rho_{11}(T)$  given  $\rho(0) = \rho_0$ .

In the article by Han et al [49], a neural network algorithm was devised for computing the ground state to the time independent Schrodinger equation (6). The Hamiltonian operator corresponds to the Born-Oppenheimer approximation [17] with  $N$  electrons and  $M$  ions:

$$\vec{r} = (\vec{r}_1, \dots, \vec{r}_N) \quad \vec{R} = (\vec{R}_1, \dots, \vec{R}_M) \quad (16)$$

$$H(\vec{r}, \vec{R}) = -\frac{1}{2} \sum_{i=1}^N \nabla_i^2 + \sum_{i=1}^N \sum_{j=i+1}^N \frac{1}{|\vec{r}_i - \vec{r}_j|} - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I}{|\vec{r}_i - \vec{R}_I|} + \sum_{I=1}^M \sum_{J=I+1}^M \frac{Z_I Z_J}{|\vec{R}_I - \vec{R}_J|} \quad (17)$$

It is assumed that the first  $N_\uparrow$  electrons are spin-up and the remaining  $N - N_\uparrow$  electrons are spin down[38]. The trial wave functions are assumed in [49] to be real valued, anti-symmetric, and have the following form:

$$\psi(\vec{r}, \vec{R}) = S(\vec{r}, \vec{R}) A^\uparrow(\vec{r}^\uparrow) A^\downarrow(\vec{r}^\downarrow) \quad (18)$$

[49] represent  $S, A^\uparrow, A^\downarrow$  using a deep neural network. The weights and biases of the network are trained in order to minimize:

$$E[\psi] = \frac{\psi^*(\vec{r}, \vec{R}) H \psi(\vec{r}, \vec{R}) d\vec{r}}{\psi^*(\vec{r}, \vec{R}) \psi(\vec{r}, \vec{R}) d\vec{r}}. \quad (19)$$

[49] represent the network parameters at epoch  $t$  as,

$$\vec{\Theta}, \quad (20)$$

and the cost function in terms of  $\vec{\Theta}$  is given as,

$$E[\psi] \equiv \Omega_{E_{ref}}(\vec{\Theta}). \quad (21)$$

The stochastic gradient descent method with learning rate  $\eta$  is used to minimize (21):

$$\Theta_{t+1} = \Theta_t - \eta \nabla_{\vec{\Theta}} \Omega_{E_{ref}}(\vec{\Theta}) \quad (22)$$

[49] report results within two percent error for  $H_2$  (two electrons),  $Be_4$  (four electrons),  $LiH_4$  (four electrons),  $He$  (two electrons), and  $H_{10}$  (ten electrons).

[53] developed a similar method as [49] for computing the ground state to the time independent Schrodinger equation (6). Both [53] and [49] express the trial wave functions using a neural network:

$$\psi(\vec{r}, \vec{R}) \approx \psi_{\vec{\Theta}}(\vec{r}) \quad (23)$$

[53] report results within three percent error for  $H_2$  (two electrons),  $Be_4$  (four electrons),  $LiH_4$  (four electrons), and  $B$  (five electrons).

1. Günther et al (2021) [48], “Quandary: An open-source C++ package for high-performance optimal control of open quantum systems.” the article by Lidar (2019) [82] (“Lecture notes on the theory of open quantum systems”) gives the derivation of the model for open quantum systems.
2. Günther and Petersson (2023) [47], “A practical approach to determine minimal quantum gate durations using amplitude-bounded quantum controls.”
3. Fei et al (2025) [32], “Switching time optimization for binary quantum optimal control.”
4. Fei et al (2023) [33], “Binary control pulse optimization for quantum systems.”
5. Kanai et al (2024) [67], “Single-Electron Qubits Based on Quantum Ring States on Solid Neon Surface.”
6. Andrade et al (2022) [3], “Engineering an effective three-spin Hamiltonian in trapped-ion systems for applications in quantum simulation.”
7. He et al (2024) [52], “Efficient Optimal Control of Open Quantum Systems”
8. Hangleiter et al (2024) [50], “Robustly learning the Hamiltonian dynamics of a superconducting quantum processor.”
9. Sun and Galperin (2025) [123], “Control of open quantum systems: The nonequilibrium Green’s function perspective.”
10. Lin Lin (2025) [83], “Dissipative preparation of many-body quantum states: Toward practical quantum advantage”
11. Kodali and Motamarri (2025) [77], “Finite-element methods for noncollinear magnetism and spin-orbit coupling in real-space pseudopotential density functional theory”
12. Barrera et al (2025) [8], “The Moving Born-Oppenheimer Approximation”
13. Nieminen et al (2023) [101], “Atomistic modeling of a superconductor–transition metal dichalcogenide–superconductor Josephson junction”
14. Mucciolo et al (2025) [95], “Green’s Function Methods for Computing Supercurrents in Josephson Junctions”
15. Williams-Young et al (2023) [130], “Distributed memory, GPU accelerated Fock construction for hybrid, Gaussian basis density functional theory”
16. Dat-Thuc Nguyen and Vo Anh Khoa (2025) [98], “An inversion approach to reconstruct the complex potential and intensity function in the Schrödinger evolution equation”
17. Meschede et al (2023) [93], “Quantum scattering of spinless particles in Riemannian manifolds”
18. Atanasov and Dandolo (2007) [5], “Curvature induced quantum potential on deformed surfaces”
19. Ferrari and Cuoghi (2008) [35], “Schrödinger equation for a particle on a curved surface in an electric and magnetic field”

### 3. The Schrodinger inverse problem

I update my process for inverse problem with potential reconstructions and NLS equation.

The time-dependent Schrödinger equation appeared in many studies of particle evolution in quantum dynamics of the systems of electrons, atoms, or molecules. The interactions of atom-photon in quantum optics and potential control to achieve a desired quantum state in quantum control problem are two significant applications in engineering. In the meantime, the inverse coefficient problem for Schrödinger equation arose but only a few numbers of works for reconstructing the complex-valued potentials recorded.

Some of the very first publications in inverse problem for Schrödinger equation were to consider and propose an approach for 1D real-valued potentials. One of those research articles shows that a set of real initial data, Dirichlet boundary data, and partial Neumann data can uniquely and stably rebuild the real-valued spatial potential in one dimensional domain. Later, it was enhanced to allow potential reconstructions from less boundary data and extend to discontinuous principal coefficient. In addition, by measuring a single sub-boundary data, one can also reconstruct spatially real-valued potentials. Multiple-coefficient including two time-independent coefficients and potentials reconstructions in the Schrödinger evolution equation have been well studied. Particularly, electric potential and potential's support are uniquely recovered from a single dynamical data.

The magnetic Schrödinger equation has also been a spotlight in the field of inverse problem where two of the first publications in this direction claimed the stable dependence of the compactly supported magnetic field and the electric potential on the Dirichlet-to-Neumann map. The results were based on the uniqueness of modulo a gauge transform and the so-called optics geometric method. Afterward, this Dirichlet-to-Neumann map approach was expanded to a nonlinear quadratic case and to solve the time-dependent potential in a periodic medium of the non-magnetic inverse problem.

Although there is a huge number of publications regarding to real-valued potential in the time-dependent Schrödinger equation for single-coefficients and multi-coefficients inversion approaches, articles for complex-valued potentials are rare. As far as we concern, the year 2019 saw one of the first mathematical publications that ones can determine the complex-valued electric potential via finitely many partial measurements on Neumann boundary within the entire time domain of the wave-function. Later, this approach was proven for the complex electromagnetic potential reconstruction in two-state magnetic Schrödinger system. Moreover, a physics-informed neural networks (PINNs) was designed to access the inversion of the 1D and 2D potential's parameters. This approach was based on the full Dirichlet-to-Neumann map and measurements on the whole boundary. PINNs then was extended for the entire potential reconstruction in saturable nonlinear Schrödinger equations. Another powerful method named Regularized-Newton-GMRES have also been developed for the complex potential in general nonlinear Schrödinger equation.

From that, I think there will be at least 15 papers needed to be add. I will add more details on this, with formula, methods,...

## 4. Quantum algorithms and Quantum emulators for benchmarking quantum computers

### 4.1. Quantum Emulators

The development of Quantum Emulators is essential for benchmarking quantum computers since (i) the level of noise in Quantum Emulators can be prescribed exactly thereby enabling practitioners to quantify the effect of noise on a given quantum circuit design, (ii) one can prescribe a given noise model[84] and then compare emulator results with an actual quantum computer in order to characterise the actual noise present in a quantum computer, and (iii) one can isolate quantum algorithmic problems from intrinsic quantum computer noise problems.

A Quantum algorithm consists of (i) state preparation and measurement and (ii) a sequence of (basic) unitary operation(s).

#### 4.1.1. state preparation and measurement

##### 1. State Preparation.

Given  $n$  qubits, there will be  $N = 2^n$  basis states. For example if  $n = 3$ , the standard basis is

$$\begin{aligned} |000\rangle &= |0\rangle \otimes |0\rangle \otimes |0\rangle \equiv \vec{0} \\ |001\rangle &= |0\rangle \otimes |0\rangle \otimes |1\rangle \equiv \vec{1} \\ &\dots \\ |111\rangle &= |1\rangle \otimes |1\rangle \otimes |1\rangle \equiv \vec{7} \end{aligned}$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

In general if one wants to prepare the state,

$$|\vec{\psi}\rangle \quad |\vec{\psi}\rangle \in C^N,$$

on a quantum computer, then one searches for as sparse a unitary matrix  $U_\psi$  as possible such that,

$$U_\psi |\vec{0}\rangle = |\vec{\psi}\rangle,$$

or, equivalently,

$$U_\psi |\vec{0}\rangle = \frac{e^{i\theta}}{||\psi||} \sum_{i_1 i_2 \dots i_n} \psi_{i_1 i_2 \dots i_n} |i_1 i_2 \dots i_n\rangle.$$

The indices  $i_k$  take values in 0, 1,  $\theta \in [0, 2\pi)$ , and

$$||\psi||^2 = \sum_{i_1 i_2 \dots i_n} |\psi_{i_1 i_2 \dots i_n}|^2$$

In preparing an  $n$  qubit quantum state, if no ancillary qubits are to be used, then the circuit depth is  $O(2^n)$  where the circuit depth is the number of “basic” quantum computer operations that cannot be performed in parallel. As reported in [60, 103], the lower and upper bounds on the number of CNOT gates required for state preparation of  $n$  qubits, with no additional ancillaries, is  $(1/2)2^n$  and  $(23/24)2^n$  respectively. As reported in [135], an optimal circuit depth of  $O(n)$  can be achieved if the number of ancilla qubits is  $O(2^n)$  [124],[110] (Corollary 1.9).

The results in [124, 110, 103] describe a very important trade off. The larger the state preparation circuit depth, the higher the odds that decoherence ruins the prepared state, on the other hand, the larger the number of ancillary qubits, the larger the quantum computer that must be built. The quantum singular value decomposition of a dense matrix [111] is an example in which the trade-off between circuit depth and number of ancillar qubits must be considered.

## 2. Measurement.

On a  $n$  qubit system, if one wants to make a measurement of the state  $|\psi\rangle$ ,

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle, \quad (24)$$

with respect to the standard basis then one can proceed as follows:

1. Define a complete set of projection operators  $P_j$ ,  $j = 0, \dots, J - 1$ ; e.g.,

$$P_j = |j\rangle \langle j| \quad J = 2^n. \quad (25)$$

It is required that,

$$\sum_{j=0}^{J-1} P_j = I.$$

- The probability of getting an outcome  $j$  on measuring the observable  $P_j$  is given by,

$$p(j) = \langle \psi | P_j | \psi \rangle,$$

and the normalized state after the measurement is given by,

$$|\psi'\rangle = \frac{P_j |\psi\rangle}{\sqrt{\langle \psi | P_j | \psi \rangle}}. \quad (26)$$

In order to make a measurement in an arbitrary basis[121], then one can proceed as follows:

- create a unitary matrix  $U$  with the arbitrary basis vectors,

$$|u_0\rangle, \dots, |u_{2^n-1}\rangle,$$

as columns.

- Apply  $U^\dagger$ :

$$|\tilde{\psi}\rangle = U^\dagger |\psi\rangle.$$

The coefficients with respect to the new basis are now the coefficients of  $|\tilde{\psi}\rangle$  with respect to the standard basis:

$$|\tilde{\psi}\rangle = \sum_{i=0}^{2^n-1} \beta_i |i\rangle \quad |\psi\rangle = \sum_{i=0}^{2^n-1} \beta_i |u_i\rangle$$

- Measure  $|\tilde{\psi}\rangle$  using the same steps as if one was making a measurement with respect to the standard basis, except that one must multiply the measured state (26) by  $U$ :

$$|\psi'\rangle = \frac{U P_j |\tilde{\psi}\rangle}{\sqrt{\langle \tilde{\psi} | P_j | \tilde{\psi} \rangle}}. \quad (27)$$

Another way to think about making a measurement in an arbitrary basis is to first define the projectors,

$$P_j \equiv |u_j\rangle \langle u_j| \quad j = 0, \dots, 2^n - 1. \quad (28)$$

Then, one measures  $|\psi\rangle$  with respect to the projectors in (28) which is equivalent to measuring  $|\tilde{\psi}\rangle$  with respect to the standard basis and then multiplying the result by  $U$ .

One can define an ‘‘Observable’’ as follows,

$$A = \lambda_0 P_0 + \lambda_1 P_1 + \dots + \lambda_{2^n-1} P_{2^n-1} \quad (29)$$

where the eigenvalues of  $A$  are  $\lambda_j$  and the corresponding eigenvectors of  $A$  are  $|u_j\rangle$ . As quoted from [121], ‘‘when we measure an observable  $A$  in a state  $|\psi\rangle$ , each measurement yields exactly one of the eigenvalues  $\lambda_j$ . The probability of obtaining  $\lambda_j$  is,

$$p(j) = \langle \psi | P_j | \psi \rangle. \quad (30)$$

Upon measurement, the state ‘‘collapses’’ to a new state

$$\frac{P_j |\psi\rangle}{\sqrt{\langle \psi | P_j | \psi \rangle}}. \quad (31)$$

The denominator is the normalization factor to account for the unit normal requirement of a quantum state.’’

A very fundamental benchmark test is to repeatedly (a) prepare a given state, and (b) measure the state with respect to an observable  $A$ . In what follows, many quantum emulators are compared and contrasted with respect to their (i) state preparation algorithm, (ii) noise modeling, (iii) parallel capability, (iv) vectorizable capability, and (v) ability to test quantum error correction algorithms for a given noise level.

The IBM Qiskit emulator[62] uses the algorithm described in Iten et al[60] for state preparation. First it was reported in [60] that for the state preparation of  $n$  qubits, a lower bound on the number of CNOT gates required is  $2^n/2$  and an upper bound is  $(23/24)2^n$ [103]. In Appendix A5 of [60], optimized state preparation algorithms were described for the state preparation of  $n = 2$  and  $n = 3$  qubits[137]; for  $n \geq 4$ , the algorithm from [103] is used.

The following state preparation algorithms have been cited with respect to the Intel Quantum simulator [140], [45], [117], [116]. Furthermore, the following noise models are available on the Intel Quantum Simulator [45]: [9], [113].

1. (IBM) “Quantum Computing with Qiskit” (2024) [62]
2. Wille et al (2018) [129], “Computer-aided design for quantum computation.”
3. “Intel Quantum Simulator: a cloud-ready high-performance simulator of quantum circuits” (2020) [45]
4. (Microsoft) “LIQUid: A Software design architecture and domain specific language for quantum computing” (2014) [128]
5. “QuESTLink—Mathematica embiggened by a hardware-optimised quantum emulator” (2020) [66]
6. (Matlab) “QCLAB: A Matlab Toolbox for Quantum Computing” (2025) [69]
7. “PennyLane-Lightning MPI: A massively scalable quantum circuit simulator based on distributed computing in CPU clusters” (2025) [68]
8. (Google) “qsim”(2020) [125]
9. (nVidia) “cuQuantum SDK: A High-Performance Library for Accelerating Quantum Science” (2023) [11]
10. “Numerical recipes in quantum information theory and quantum computing: an adventure in FORTRAN 90”(2021) [108]
11. Nguyen et al(2022) [99] “Tensor network quantum virtual machine for simulating quantum circuits at exascale”
12. Zhang et al (2024) [134], “Quantum circuit simulation with fast tensor decision diagram”
13. Viamontes et al (2004) [127], “High-performance QuIDD-based simulation of quantum circuits.”
14. Ito and Yoshioka (2024) [61], “Quantum computer simulation on HPC.”
15. Xu et al (2024) [132], “Atlas: Hierarchical partitioning for quantum circuit simulation on gpus”
16. Zhang et al (2021) [133], “HyQuas: hybrid partitioner based quantum circuit simulation system on GPU”
17. Li et al (2021) [81], “Sv-sim: scalable pgas-based state vector simulation of quantum circuits”
18. Kolarovszki et al (2025) [78], “Piquasso: A photonic quantum computer simulation software platform”
19. Lotshaw et al (2023) [84], “Modeling noise in global Molmer-Sorensen interactions applied to quantum approximate optimization.”
20. De Raedt et al (2007) [26], “Massively parallel quantum computer simulator”
21. De Raedt et al (2019) [26], “Massively parallel quantum computer simulator, eleven years later.”
22. Amazon Braket

## 4.2. Quantum Algorithms

1. Sawaya et al (2023) [112], “HamLib: A Library of Hamiltonians for Benchmarking Quantum Algorithms and Hardware.”
2. Boixo et al (2018) [15], “Characterizing quantum supremacy in near-term devices”
3. Kitaev (1995) [73], “Quantum measurements and the Abelian stabilizer problem” (quantum phase estimation)
4. Shor (1999) [120], “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”
5. Callison and Chancellor (2022) [21] “Hybrid quantum-classical algorithms in the noisy intermediate-scale quantum era and beyond”
6. Jin et al (2024) [63], “Quantum Simulation for Quantum Dynamics with Artificial Boundary Conditions.”
7. S. Jin, N. Liu, and Y. Yu (2024) [65], “Quantum Circuits for the heat equation with physical boundary conditions via Schrodingerisation”
8. Cerezo et al (2021) [22] “Variational quantum algorithms”
9. Lotshaw et al (2021) [86] “Simulations of frustrated Ising Hamiltonians using quantum approximate optimization”
10. Nielsen and Chuang (2010) [100] “Quantum computation and quantum information” (quantum Fourier Transform and quantum phase estimation)
11. Ding and Lin (2023) [29] “Simultaneous estimation of multiple eigenvalues with short-depth quantum circuit on early fault-tolerant quantum computers”
12. Biamonte et al (2017) [13] “Quantum machine learning”
13. Benedetti et al (2019)[12] “Parameterized quantum circuits as machine learning models”
14. Khait et al(2023) [70] “Variational quantum eigensolvers in the era of distributed quantum computers”
15. Sawaya et al (2023) [112], “HamLib: A Library of Hamiltonians for Benchmarking Quantum Algorithms and Hardware”

16. Georgiadou et al(2024) [43] Hele-Shaw simulation on quantum computer.
17. Kim et al (2023) [72], “Evidence for the utility of quantum computing before fault tolerance”
18. Zoufal et al (2019) [139], “Quantum generative adversarial networks for learning and loading random distributions”
19. Nghiem and Tzu-Chieh (2023) [97], “Quantum algorithm for estimating largest eigenvalues”
20. Cadi et al (2024) [20], “Folded spectrum vqe: A quantum computing method for the calculation of molecular excited states”
21. Tilly et al (2022) [126], “The variational quantum eigensolver: a review of methods and best practices”
22. Qing and Xie (2023) [106], “Use VQE to calculate the ground energy of hydrogen molecules on IBM Quantum”
23. Ryu et al (2021) [111], “A Variational Quantum Algorithm for Ordered SVD”
24. Shirakawa et al (2024) [118], “Automatic quantum circuit encoding of a given arbitrary quantum state”
25. Harrow et al (2009) [51], “Quantum algorithm for linear systems of equations”
26. Daribayev et al (2023) [25], “Implementation of the hhl algorithm for solving the poisson equation on quantum simulators.”
27. Bravo-Prieto et al (2023) [18], “Variational quantum linear solver”
28. Gilyen et al (2019) [41], “Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics”
29. Gundlach et al (2025) [46], “Quantum Advantage in Computational Chemistry?”
30. Ko et al (2023) [76], “Implementation of the density-functional theory on quantum computers with linear scaling with respect to the number of atoms”
31. Jin et al (2025) [64], “Quantum preconditioning method for linear systems problems via Schrodingerization”
32. Itani et al (2024) [59], “Quantum algorithm for lattice Boltzmann (QALB) simulation of incompressible fluids with a nonlinear collision term”
33. Hibat-Allah et al (2024) [55], “A framework for demonstrating practical quantum advantage: comparing quantum against classical generative models”
34. Herrmann et al (2023) [54], “Quantum utility—definition and assessment of a practical quantum advantage”
35. Di Bartolomeo et al (2023) [28], “Noisy gates for simulating quantum computers.”
36. De Luo et al (2025) [87], “Quantum simulation of bubble nucleation across a quantum phase transition.”
37. Maciejewski et al (2024) [88], “Improving Quantum Approximate Optimization by Noise-Directed Adaptive Remapping.”
38. Seki et al (2025) [114], “Simulating Floquet scrambling circuits on trapped-ion quantum computers”
39. Rakyta et al (2024) [107], “Highly optimized quantum circuits synthesized via data-flow engines”
40. Popovici et al (2025) [104], “A Closeness Centrality-based Circuit Partitioner for Quantum Simulations”
41. Hsu et al (2024) [57], “Toward cost-effective quantum circuit simulation with performance tuning techniques”
42. Park et al (2022) [102], “SnuQS: scaling quantum circuit simulation using storage devices”

## 5. Quantum Error Correction algorithms and surface codes

1. Shor (1995) [119], “Scheme for reducing decoherence in quantum computer memory.”
2. Anthony-Petersen et al (2024) [4], “A stress-induced source of phonon bursts and quasiparticle poisoning.”
3. Lotshaw et al (2024) [85], “Exactly solvable model of light-scattering errors in quantum simulations with metastable trapped-ion qubits.”
4. Kubischta, Eric and Teixeira, Ian (2023) [80], “Family of quantum codes with exotic transversal gates.”
5. Barends et al (2014) [7], “Superconducting quantum circuits at the surface code threshold for fault tolerance.”
6. Baum et al (2021) [10], “Experimental deep reinforcement learning for error-robust gate-set design on a superconducting quantum computer.”
7. Cheng and Guo (2025) [23], “Modeling correlated-noise in silicon spin qubit device.”
8. Putterman et al (2025) [105], “Hardware-efficient quantum error correction via concatenated bosonic qubits”
9. Knill and Laflamme (1997) [75], “Theory of quantum error-correcting codes”
10. Eastin and Knill (2009) [30], “Restrictions on transversal encoded quantum gate sets”

11. Ide et al (2025) [58], “Fault-tolerant logical measurements via homological measurement”
12. Bravyi and Kitaev (2002) [19], “Fermionic quantum computation”
13. Dennis et al (2002) [27], “Topological quantum memory”
14. Kitaev (2003) [74], “Fault-tolerant quantum computation by anyons”
15. Roffe (2019) [109], “Quantum error correction: an introductory guide”
16. Bluvstein et al (2024) [14], “Logical quantum processor based on reconfigurable atom arrays”
17. Marra (2022) [92], “Majorana nanowires for topological quantum computation”
18. L. Hormozi (2007) [56], “Topological Quantum Compiling”
19. Evered et al (2023) [31], “High-fidelity parallel entangling gates on a neutral-atom quantum computer”
20. Wintersperger et al (2023) [131], “Neutral atom quantum computing hardware: performance and end-user perspective”
21. Gonthier et al (2022) [42], “Measurements as a roadblock to near-term practical quantum advantage in chemistry: Resource analysis”
22. Dalton et al (2024) [24], “Quantifying the effect of gate errors on variational quantum eigensolvers for quantum chemistry”
23. Nautrup et al (2019) [96], “Optimizing quantum error correction codes with reinforcement learning”
24. Krinner et al (2022) [79], “Realizing repeated quantum error correction in a distance-three surface code”
25. Stetina and Wiebe (2025) [122], “First-Quantized Quantum Simulation of Non-Relativistic QED with Emergent Topologically Protected Coulomb Interactions”
26. Zou and Hoi-Kwong (2025) [138], “Algebraic Topology Principles behind Topological Quantum Error Correction”
27. Google Quantum (2025) [1], “Quantum error correction below the surface code threshold”
28. Khan et al (2025) [71], “Separate and efficient characterization of state-preparation and measurement errors using single-qubit operations”
29. Bonilla et al (2021) [16], “The XZZX surface code”
30. Aasen et al (2025) [2], “Roadmap to fault tolerant quantum computation using topological qubit arrays”
31. K. Fujii (2015) [39], “Quantum Computation with Topological Codes: from qubit to topological fault-tolerance”

## 6. Bibliography

Two bibliographic style files (\*.bst) are provided — model1-num-names.bst and model2-names.bst — the first one can be used for the numbered scheme. This can also be used for the numbered with new options of natbib.sty. The second one is for the author year scheme. When you use model2-names.bst, the citation commands will be like \citep, \citet, \citealt etc. However when you use model1-num-names.bst, you may use only \cite command. The bibliography environment. Each reference is a \bibitem and each \bibitem is identified by a label, by which it can be cited in the text:

In connection with cross-referencing and possible future hyperlinking it is not a good idea to collect more than one literature item in one \bibitem. The so-called Harvard or author-year style of referencing is enabled by the L<sup>A</sup>T<sub>E</sub>X package natbib. With this package the literature can be cited as follows:

- Parenthetical: \citep{WB96} produces (Wettig & Brown, 1996).
- Textual: \citet{ESG96} produces Elson et al. (1996).
- An affix and part of a reference:  
\citep[e.g.] [Ch. 2]{Gea97} produces (e.g. Governato et al., 1997, Ch. 2).

In the numbered scheme of citation, \cite{<label>} is used, since \citep or \citet has no relevance in the numbered scheme. natbib package is loaded by cas-sc with numbers as default option. You can change this to author-year or harvard scheme by adding option authoryear in the class loading command. If you want to use more options of the natbib package, you can do so with the \biboptions command. For details of various options of the natbib package, please take a look at the natbib documentation, which is part of any standard L<sup>A</sup>T<sub>E</sub>X installation.

## A. My Appendix

Appendix sections are coded under `\appendix`.

`\printcredits` command is used after appendix sections to list author credit taxonomy contribution roles tagged using `\credit` in frontmatter.

## References

- [1] , 2025. Quantum error correction below the surface code threshold. *Nature* 638, 920–926.
- [2] Aasen, D., Aghaee, M., Alam, Z., Andrzejczuk, M., Antipov, A., Astafev, M., Avilovas, L., Barzegar, A., Bauer, B., Becker, J., Bello-Rivas, J.M., Bhaskar, U., Bocharov, A., Boddapati, S., Bohn, D., Bommer, J., Bonderson, P., Borovsky, J., Bourdet, L., Boutin, S., Brown, T., Campbell, G., Casparis, L., Chakravarthi, S., Chao, R., Chapman, B.J., Chatoor, S., Christensen, A.W., Codd, P., Cole, W., Cooper, P., Corsetti, F., Cui, A., van Dam, W., Dandachi, T.E., Daraeizadeh, S., Dumitrascu, A., Ekeffjård, A., Fallahi, S., Galletti, L., Gardner, G., Gatta, R., Gavranovic, H., Goulding, M., Govender, D., Griggio, F., Grigoryan, R., Grijalva, S., Gronin, S., Gukelberger, J., Haah, J., Hamdast, M., Hansen, E.B., Hastings, M., Heedt, S., Ho, S., Hogaboam, J., Holgaard, L., Hoogdalem, K.V., Indrapiomkul, J., Ingerslev, H., Ivancevic, L., Jablonski, S., Jensen, T., Jhoja, J., Jones, J., Kalashnikov, K., Kallaheer, R., Kalra, R., Karimi, F., Karzig, T., Kimes, S., Kliuchnikov, V., Kloster, M.E., Knapp, C., Knee, D., Koski, J., Kostamo, P., Kuesel, J., Lackey, B., Laeven, T., Lai, J., de Lange, G., Larsen, T., Lee, J., Lee, K., Leum, G., Li, K., Lindemann, T., Lucas, M., Lutchyn, R., Madsen, M.H., Madulid, N., Manfra, M., Markussen, S.B., Martinez, E., Mattila, M., Mattinson, J., McNeil, R., Mei, A.R., Mishmash, R.V., Mohandas, G., Mollgaard, C., de Moor, M., Morgan, T., Moussa, G., Narla, A., Nayak, C., Nielsen, J.H., Nielsen, W.H.P., Nolet, F., Nystrom, M., O'Farrell, E., Otani, K., Paetznick, A., Papon, C., Paz, A., Petersson, K., Petit, L., Pikulin, D., Pons, D.O.F., Quinn, S., Rajpalke, M., Ramirez, A.A., Rasmussen, K., Razmadze, D., Reichardt, B., Ren, Y., Renner, K., Riccomini, R., Sadovskyy, I., Sainiemi, L., Saldaña, J.C.E., Sanlorenzo, I., Schaal, S., Schmidgall, E., Sfiligoj, C., da Silva, M.P., Singh, S., Sinha, S., Soeken, M., Sohr, P., Stankevicius, T., Stek, L., Strøm-Hansen, P., Stuppard, E., Sundaram, A., Suominen, H., Suter, J., Suzuki, S., Svore, K., Teicher, S., Thiagarajah, N., Tholapi, R., Thomas, M., Tom, D., Toomey, E., Tracy, J., Troyer, M., Turley, M., Turner, M.D., Upadhyay, S., Urban, I., Vashchillo, A., Viazmitinov, D., Vogel, D., Wang, Z., Watson, J., Webster, A., Weston, J., Williamson, T., Winkler, G.W., van Woerkom, D.J., Wütz, B.P., Yang, C.K., Yu, R., Yucelen, E., Zamorano, J.H., Zeisel, R., Zheng, G., Zilke, J., Zimmerman, A., 2025. Roadmap to fault tolerant quantum computation using topological qubit arrays. URL: <https://arxiv.org/abs/2502.12252>, [arXiv:2502.12252](https://arxiv.org/abs/2502.12252).
- [3] Andrade, B., Davoudi, Z., Graß, T., Hafezi, M., Pagano, G., Seif, A., 2022. Engineering an effective three-spin hamiltonian in trapped-ion systems for applications in quantum simulation. *Quantum Science and Technology* 7, 034001. URL: <https://dx.doi.org/10.1088/2058-9565/ac5f5b>, doi:10.1088/2058-9565/ac5f5b.
- [4] Anthony-Petersen, R., Biekert, A., Bunker, R., Chang, C.L., Chang, Y.Y., Chaplinsky, L., Fascione, E., Fink, C.W., Garcia-Sciveres, M., Germond, R., et al., 2024. A stress-induced source of phonon bursts and quasiparticle poisoning. *Nature Communications* 15, 6444.
- [5] Atanasov, V., Dandoloﬀ, R., 2007. Curvature induced quantum potential on deformed surfaces. *Physics Letters A* 371, 118–123.
- [6] Bao, W., Jin, S., Markowich, P.A., 2002. On time-splitting spectral approximations for the schrödinger equation in the semiclassical regime. *Journal of Computational Physics* 175, 487–524. URL: <https://www.sciencedirect.com/science/article/pii/S0021999101969566>, doi:<https://doi.org/10.1006/jcph.2001.6956>.
- [7] Barends, R., Kelly, J., Megrant, A., Veitia, A., Sank, D., Jeffrey, E., White, T.C., Mutus, J., Fowler, A.G., Campbell, B., et al., 2014. Superconducting quantum circuits at the surface code threshold for fault tolerance. *Nature* 508, 500–503.
- [8] Barrera, B., Arovas, D.P., Chandran, A., Polkovnikov, A., 2025. The moving born-oppenheimer approximation. URL: <https://arxiv.org/abs/2502.17557>, [arXiv:2502.17557](https://arxiv.org/abs/2502.17557).
- [9] Bassi, A., Deckert, D.A., 2008. Noise gates for decoherent quantum circuits. *Physical Review A—Atomic, Molecular, and Optical Physics* 77, 032323.
- [10] Baum, Y., Amico, M., Howell, S., Hush, M., Liuzzi, M., Mundada, P., Merkh, T., Carvalho, A.R., Biercuk, M.J., 2021. Experimental deep reinforcement learning for error-robust gate-set design on a superconducting quantum computer. *PRX Quantum* 2, 040324.
- [11] Bayraktar, H., Charara, A., Clark, D., Cohen, S., Costa, T., Fang, Y.L.L., Gao, Y., Guan, J., Gunnels, J., Haidar, A., Hehn, A., Hohnerbach, M., Jones, M., Lubowe, T., Lyakh, D., Morino, S., Springer, P., Stanwyck, S., Terentyev, I., Varadhan, S., Wong, J., Yamaguchi, T., 2023. cuquantum sdk: A high-performance library for accelerating quantum science, in: 2023 IEEE International Conference on Quantum Computing and Engineering (QCE), pp. 1050–1061. doi:10.1109/QCE57702.2023.00119.
- [12] Benedetti, M., Lloyd, E., Sack, S., Fiorentini, M., 2019. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology* 4, 043001. URL: <https://dx.doi.org/10.1088/2058-9565/ab4eb5>, doi:10.1088/2058-9565/ab4eb5.
- [13] Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., Lloyd, S., 2017. Quantum machine learning. *Nature* 549, 195–202.
- [14] Bluvstein, D., Evered, S.J., Geim, A.A., Li, S.H., Zhou, H., Manóvitz, T., Ebadi, S., Cain, M., Kalinowski, M., Hangleiter, D., et al., 2024. Logical quantum processor based on reconfigurable atom arrays. *Nature* 626, 58–65.
- [15] Boixo, S., Isakov, S.V., Smelyanskiy, V.N., Babbush, R., Ding, N., Jiang, Z., Bremner, M.J., Martinis, J.M., Neven, H., 2018. Characterizing quantum supremacy in near-term devices. *Nature Physics* 14, 595–600.
- [16] Bonilla Ataides, J.P., Tuckett, D.K., Bartlett, S.D., Flammia, S.T., Brown, B.J., 2021. The xxxz surface code. *Nature communications* 12, 2172.
- [17] Born, M., Oppenheimer, R., 1927. Zur quantentheorie der molekeln. *Annalen der Physik* 389, 457–484. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/andp.19273892002>, doi:<https://doi.org/10.1002/andp.19273892002>, [arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/andp.19273892002](https://onlinelibrary.wiley.com/doi/pdf/10.1002/andp.19273892002).
- [18] Bravo-Prieto, C., LaRose, R., Cerezo, M., Subasi, Y., Cincio, L., Coles, P.J., 2023. Variational quantum linear solver. *Quantum* 7, 1188.
- [19] Bravyi, S.B., Kitaev, A.Y., 2002. Fermionic quantum computation. *Annals of Physics* 298, 210–226.

- [20] Cadi Tazi, L., Thom, A.J., 2024. Folded spectrum vqe: A quantum computing method for the calculation of molecular excited states. *Journal of Chemical Theory and Computation* 20, 2491–2504.
- [21] Callison, A., Chancellor, N., 2022. Hybrid quantum-classical algorithms in the noisy intermediate-scale quantum era and beyond. *Physical Review A* 106, 010101.
- [22] Cerezo, M., Arrasmith, A., Babbush, R., Benjamin, S.C., Endo, S., Fujii, K., McClean, J.R., Mitarai, K., Yuan, X., Cincio, L., et al., 2021. Variational quantum algorithms. *Nature Reviews Physics* 3, 625–644.
- [23] Cheng, G., Guo, J., 2025. Modeling correlated-noise in silicon spin qubit device. *APL Quantum* 2.
- [24] Dalton, K., Long, C.K., Yordanov, Y.S., Smith, C.G., Barnes, C.H., Mertig, N., Arvidsson-Shukur, D.R., 2024. Quantifying the effect of gate errors on variational quantum eigensolvers for quantum chemistry. *npj Quantum Information* 10, 18.
- [25] Daribayev, B., Mukhanbet, A., Imankulov, T., 2023. Implementation of the hhl algorithm for solving the poisson equation on quantum simulators. *Applied Sciences* 13, 11491.
- [26] De Raedt, H., Jin, F., Willsch, D., Willsch, M., Yoshioka, N., Ito, N., Yuan, S., Michielsen, K., 2019. Massively parallel quantum computer simulator, eleven years later. *Computer Physics Communications* 237, 47–61.
- [27] Dennis, E., Kitaev, A., Landahl, A., Preskill, J., 2002. Topological quantum memory. *Journal of Mathematical Physics* 43, 4452–4505.
- [28] Di Bartolomeo, G., Vischi, M., Cesa, F., Wixinger, R., Grossi, M., Donadi, S., Bassi, A., 2023. Noisy gates for simulating quantum computers. *Physical Review Research* 5, 043210.
- [29] Ding, Z., Lin, L., 2023. Simultaneous estimation of multiple eigenvalues with short-depth quantum circuit on early fault-tolerant quantum computers. *Quantum* 7, 1136.
- [30] Eastin, B., Knill, E., 2009. Restrictions on transversal encoded quantum gate sets. *Physical review letters* 102, 110502.
- [31] Evered, S.J., Bluvstein, D., Kalinowski, M., Ebadi, S., Manovitz, T., Zhou, H., Li, S.H., Geim, A.A., Wang, T.T., Maskara, N., et al., 2023. High-fidelity parallel entangling gates on a neutral-atom quantum computer. *Nature* 622, 268–272.
- [32] Fei, X., Brady, L., Larson, J., Leyffer, S., Shen, S., 2025. Switching time optimization for binary quantum optimal control. *ACM Transactions on Quantum Computing* 6, 1–38.
- [33] Fei, X., Brady, L.T., Larson, J., Leyffer, S., Shen, S., 2023. Binary control pulse optimization for quantum systems. *Quantum* 7, 892.
- [34] Feit, M., Fleck, J., Steiger, A., 1982. Solution of the schrödinger equation by a spectral method. *Journal of Computational Physics* 47, 412–433. URL: <https://www.sciencedirect.com/science/article/pii/0021999182900912>, doi:[https://doi.org/10.1016/0021-9991\(82\)90091-2](https://doi.org/10.1016/0021-9991(82)90091-2).
- [35] Ferrari, G., Cuoghi, G., 2008. Schrödinger equation for a particle on a curved surface in an electric and magnetic field. *Physical review letters* 100, 230403.
- [36] Feynman, R.P., 1982. Simulating physics with quantum computers. *International Journal of Theoretical Physics* 21, 467–488.
- [37] Feynman, R.P., 1986. Quantum mechanical computers. *Foundations of Physics* 16, 507–531. Originally published in *Optics News*, February 1985, pp. 11–20.
- [38] Foulkes, W.M., Mitas, L., Needs, R., Rajagopal, G., 2001. Quantum monte carlo simulations of solids. *Reviews of Modern Physics* 73, 33.
- [39] Fujii, K., 2015. Quantum computation with topological codes: from qubit to topological fault-tolerance. URL: <https://arxiv.org/abs/1504.01444>, arXiv:1504.01444.
- [40] Gamble, S., 2019. Quantum computing: What it is, why we want it, and how we're trying to get it, in: *Frontiers of Engineering: Reports on Leading-Edge Engineering from the 2018 Symposium*, National Academies Press. pp. 5–8.
- [41] Gilyén, A., Su, Y., Low, G.H., Wiebe, N., 2019. Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics, in: *Proceedings of the 51st annual ACM SIGACT symposium on theory of computing*, pp. 193–204.
- [42] Gonthier, J.F., Radin, M.D., Buda, C., Doskocil, E.J., Abuan, C.M., Romero, J., 2022. Measurements as a roadblock to near-term practical quantum advantage in chemistry: Resource analysis. *Physical Review Research* 4, 033154.
- [43] Gopalakrishnan Meena, M., Gottiparthi, K.C., Lietz, J.G., Georgiadou, A., Coello Pérez, E.A., 2024. Solving the hele–shaw flow using the harrow–hassidim–lloyd algorithm on superconducting devices: A study of efficiency and challenges. *Physics of Fluids* 36.
- [44] Grivopoulos, S., 2005. Optimal control of quantum systems. University of California, Santa Barbara.
- [45] Guerreschi, G.G., Hogaboam, J., Baruffa, F., Sawaya, N.P.D., 2020. Intel Quantum Simulator: a cloud-ready high-performance simulator of quantum circuits. *Quantum Science and Technology* 5. URL: <https://doi.org/10.1088/2058-9565/ab8505>, doi:10.1088/2058-9565/ab8505.
- [46] Gundlach, H., Sharkey, K., Lynch, J., Hazoglou, V., Hsu, K.C., Dukatz, C., Crane, E., Walczyk, K., Bodziak, M., Galatsanos-Dueck, J., Thompson, N., 2025. Quantum advantage in computational chemistry? URL: <https://arxiv.org/abs/2508.20972>, arXiv:2508.20972.
- [47] Günther, S., Petersson, N.A., 2023. A practical approach to determine minimal quantum gate durations using amplitude-bounded quantum controls. *AVS Quantum Science* 5.
- [48] Günther, S., Petersson, N.A., DuBois, J.L., 2021. Quandary: An open-source c++ package for high-performance optimal control of open quantum systems, in: *2021 IEEE/ACM Second International Workshop on Quantum Computing Software (QCS)*, pp. 88–98. doi:10.1109/QCS54837.2021.00014.
- [49] Han, J., Zhang, L., E, W., 2019. Solving many-electron schrödinger equation using deep neural networks. *Journal of Computational Physics* 399, 108929. URL: <https://www.sciencedirect.com/science/article/pii/S0021999119306345>, doi:<https://doi.org/10.1016/j.jcp.2019.108929>.
- [50] Hangleiter, D., Roth, I., Fuksa, J., Eisert, J., Roushan, P., 2024. Robustly learning the hamiltonian dynamics of a superconducting quantum processor. *Nature Communications* 15, 9595.
- [51] Harrow, A.W., Hassidim, A., Lloyd, S., 2009. Quantum algorithm for linear systems of equations. *Physical review letters* 103, 150502.
- [52] He, W., Li, T., Li, X., Li, Z., Wang, C., Wang, K., 2024. Efficient optimal control of open quantum systems. URL: <https://arxiv.org/abs/2405.19245>, arXiv:2405.19245.

- [53] Hermann, J., Schätzle, Z., Noé, F., 2020. Deep-neural-network solution of the electronic schrödinger equation. *Nature Chemistry* 12, 891–897.
- [54] Herrmann, N., Arya, D., Doherty, M.W., Mingare, A., Pillay, J.C., Preis, F., Prestel, S., 2023. Quantum utility—definition and assessment of a practical quantum advantage, in: 2023 IEEE International Conference on Quantum Software (QSW), IEEE, pp. 162–174.
- [55] Hibat-Allah, M., Mauri, M., Carrasquilla, J., Perdomo-Ortiz, A., 2024. A framework for demonstrating practical quantum advantage: comparing quantum against classical generative models. *Communications Physics* 7, 68.
- [56] Hormozi, L., 2007. Topological Quantum Compiling. Phd thesis. Florida State University. Tallahassee, FL.
- [57] Hsu, N.W., Wang, C.C., Hsu, C.H., Tu, C.H., Hung, S.H., 2024. Toward cost-effective quantum circuit simulation with performance tuning techniques. *Connection Science* 36, 2349541. URL: <https://doi.org/10.1080/09540091.2024.2349541>, doi:10.1080/09540091.2024.2349541, arXiv:<https://doi.org/10.1080/09540091.2024.2349541>.
- [58] Ide, B., Gowda, M.G., Nadkarni, P.J., Dauphinais, G., 2025. Fault-tolerant logical measurements via homological measurement. *Physical Review X* 15, 021088.
- [59] Itani, W., Sreenivasan, K.R., Succi, S., 2024. Quantum algorithm for lattice boltzmann (qalb) simulation of incompressible fluids with a nonlinear collision term. *Physics of Fluids* 36.
- [60] Iten, R., Colbeck, R., Kukuljan, I., Home, J., Christandl, M., 2016. Quantum circuits for isometries. *Phys. Rev. A* 93, 032318. URL: <https://link.aps.org/doi/10.1103/PhysRevA.93.032318>, doi:10.1103/PhysRevA.93.032318.
- [61] Ito, N., Yoshioka, N., 2024. Quantum computer simulation on hpc, in: Proceedings of the 21st ACM International Conference on Computing Frontiers: Workshops and Special Sessions (CF '24 Companion). Association for Computing Machinery, New York, NY, USA, 144. doi:<https://doi.org/10.1145/3637543.3653433>.
- [62] Javadi-Abhari, A., Treinish, M., Krsulich, K., Wood, C.J., Lishman, J., Gacon, J., Martiel, S., Nation, P.D., Bishop, L.S., Cross, A.W., et al., 2024. Quantum computing with qiskit. arXiv preprint arXiv:2405.08810 .
- [63] Jin, S., Li, X., Liu, N., Yu, Y., 2024a. Quantum simulation for quantum dynamics with artificial boundary conditions. *SIAM Journal on Scientific Computing* 46, B403–B421. URL: <https://doi.org/10.1137/23M1563451>, doi:10.1137/23M1563451, arXiv:<https://doi.org/10.1137/23M1563451>.
- [64] Jin, S., Liu, N., Ma, C., Yu, Y., 2025. Quantum preconditioning method for linear systems problems via schrödingerization. arXiv preprint arXiv:2505.06866 .
- [65] Jin, S., Liu, N., Yu, Y., 2024b. Quantum circuits for the heat equation with physical boundary conditions via schrodingerisation. arXiv preprint arXiv:2407.15895 .
- [66] Jones, T., Benjamin, S., 2020. Questlink—mathematica embiggened by a hardware-optimised quantum emulator\*. *Quantum Science and Technology* 5, 034012. URL: <http://dx.doi.org/10.1088/2058-9565/ab8506>, doi:10.1088/2058-9565/ab8506.
- [67] Kanai, T., Jin, D., Guo, W., 2024. Single-electron qubits based on quantum ring states on solid neon surface. *Phys. Rev. Lett.* 132, 250603. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.132.250603>, doi:10.1103/PhysRevLett.132.250603.
- [68] Kang, J.H., Ryu, H., 2025. PennyLane-lightning mpi: A massively scalable quantum circuit simulator based on distributed computing in cpu clusters. arXiv preprint arXiv:2508.13615 .
- [69] Keip, S., Camps, D., Beeumen, R.V., 2025. Qclab: A matlab toolbox for quantum computing. URL: <https://arxiv.org/abs/2503.03016>, arXiv:2503.03016.
- [70] Khait, I., Tham, E., Segal, D., Brodutch, A., 2023. Variational quantum eigensolvers in the era of distributed quantum computers. *Physical Review A* 108, L050401.
- [71] Khan, M.Q., Norris, L.M., Viola, L., 2025. Separate and efficient characterization of state-preparation and measurement errors using single-qubit operations. arXiv preprint arXiv:2509.19448 .
- [72] Kim, Y., Eddins, A., Anand, S., Wei, K.X., Van Den Berg, E., Rosenblatt, S., Nayfeh, H., Wu, Y., Zaletel, M., Temme, K., et al., 2023. Evidence for the utility of quantum computing before fault tolerance. *Nature* 618, 500–505.
- [73] Kitaev, A.Y., 1995. Quantum measurements and the abelian stabilizer problem. arXiv preprint quant-ph/9511026 .
- [74] Kitaev, A.Y., 2003. Fault-tolerant quantum computation by anyons. *Annals of physics* 303, 2–30.
- [75] Knill, E., Laflamme, R., 1997. Theory of quantum error-correcting codes. *Physical Review A* 55, 900.
- [76] Ko, T., Li, X., Wang, C., 2023. Implementation of the density-functional theory on quantum computers with linear scaling with respect to the number of atoms. arXiv preprint arXiv:2307.07067 .
- [77] Kodali, N., Motamarri, P., 2025. Finite-element methods for noncollinear magnetism and spin-orbit coupling in real-space pseudopotential density functional theory. *Phys. Rev. B* 111, 195129. URL: <https://link.aps.org/doi/10.1103/PhysRevB.111.195129>, doi:10.1103/PhysRevB.111.195129.
- [78] Kolarovszki, Z., Rybocyte, T., Rakyt, P., Kaposi, Á., Poór, B., Józsi, S., Nagy, D.T., Varga, H., El-Safty, K.H., Morse, G., et al., 2025. Piquasso: A photonic quantum computer simulation software platform. *Quantum* 9, 1708.
- [79] Krinner, S., Lacroix, N., Remm, A., Di Paolo, A., Genois, E., Leroux, C., Hellings, C., Lazar, S., Swiadek, F., Herrmann, J., et al., 2022. Realizing repeated quantum error correction in a distance-three surface code. *Nature* 605, 669–674.
- [80] Kubischta, E., Teixeira, I., 2023. Family of quantum codes with exotic transversal gates. *Physical Review Letters* 131, 240601.
- [81] Li, A., Fang, B., Granade, C., Prawiroatmodjo, G., Heim, B., Roetteler, M., Krishnamoorthy, S., 2021. Sv-sim: scalable pgas-based state vector simulation of quantum circuits, in: Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis, pp. 1–14. doi:10.1145/3458817.3476169.
- [82] Lidar, D.A., 2019. Lecture notes on the theory of open quantum systems. arXiv preprint arXiv:1902.00967 .
- [83] Lin, L., 2025. Dissipative preparation of many-body quantum states: Toward practical quantum advantage. *APL Computational Physics* 1, 010901.
- [84] Lotshaw, P.C., Battles, K.D., Gard, B., Buchs, G., Humble, T.S., Herold, C.D., 2023a. Modeling noise in global mølmer-sørensen interactions applied to quantum approximate optimization. *Phys. Rev. A* 107, 062406. URL: <https://link.aps.org/doi/10.1103/PhysRevA.107.062406>, doi:10.1103/PhysRevA.107.062406.

- [85] Lotshaw, P.C., Sawyer, B.C., Herold, C.D., Buchs, G., 2024. Exactly solvable model of light-scattering errors in quantum simulations with metastable trapped-ion qubits. *Phys. Rev. A* 110, L030803. URL: <https://link.aps.org/doi/10.1103/PhysRevA.110.L030803>, doi:10.1103/PhysRevA.110.L030803.
- [86] Lotshaw, P.C., Xu, H., Khalid, B., Buchs, G., Humble, T.S., Banerjee, A., 2023b. Simulations of frustrated ising hamiltonians using quantum approximate optimization. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 381, 20210414. URL: <https://royalsocietypublishing.org/doi/abs/10.1098/rsta.2021.0414>, doi:10.1098/rsta.2021.0414, arXiv:<https://royalsocietypublishing.org/doi/pdf/10.1098/rsta.2021.0414>.
- [87] Luo, D., Surace, F.M., De, A., Lerose, A., Bennewitz, E.R., Ware, B., Schuckert, A., Davoudi, Z., Gorshkov, A.V., Katz, O., Monroe, C., 2025. Quantum simulation of bubble nucleation across a quantum phase transition. URL: <https://arxiv.org/abs/2505.09607>, arXiv:2505.09607.
- [88] Maciejewski, F.B., Biamonte, J., Hadfield, S., Venturelli, D., 2024. Improving quantum approximate optimization by noise-directed adaptive remapping. URL: <https://arxiv.org/abs/2404.01412>, arXiv:2404.01412.
- [89] Mack, C., 2015. The multiple lives of moore's law. *IEEE Spectrum* 52, 31–31.
- [90] Manzano, D., 2020. A short introduction to the lindblad master equation. *Aip advances* 10.
- [91] Markov, I.L., 2014. Limits on fundamental limits to computation. *Nature* 512, 147–154.
- [92] Marra, P., 2022. Majorana nanowires for topological quantum computation. *Journal of Applied Physics* 132.
- [93] Meschede, L., Schwager, B., Schulz, D., Berakdar, J., 2023. Quantum scattering of spinless particles in riemannian manifolds. *Phys. Rev. A* 107, 062806. URL: <https://link.aps.org/doi/10.1103/PhysRevA.107.062806>, doi:10.1103/PhysRevA.107.062806.
- [94] Moore, G., 2021. Cramming more components onto integrated circuits (1965) .
- [95] Mucciolo, E.R., Nieminen, J., Xiao, X., Chiu, W.C., Leuenberger, M.N., Bansil, A., 2025. Green's function methods for computing supercurrents in josephson junctions. URL: <https://arxiv.org/abs/2509.07165>, arXiv:2509.07165.
- [96] Nautrup, H.P., Delfosse, N., Dunjko, V., Briegel, H.J., Friis, N., 2019. Optimizing quantum error correction codes with reinforcement learning. *Quantum* 3, 215.
- [97] Nghiem, N.A., Wei, T.C., 2023. Quantum algorithm for estimating largest eigenvalues. *Physics Letters A* 488, 129138.
- [98] Nguyen, D.T., Khoa, V.A., 2025. An inversion approach to reconstruct the complex potential and intensity function in the schrödinger evolution equation. *Journal of Computational Physics* , 114476.
- [99] Nguyen, T., Lyakh, D., Dumitrescu, E., Clark, D., Larkin, J., McCaskey, A., 2022. Tensor network quantum virtual machine for simulating quantum circuits at exascale. *ACM Transactions on Quantum Computing* 4, 1–21.
- [100] Nielsen, M.A., Chuang, I.L., 2010. Quantum computation and quantum information. Cambridge university press.
- [101] Nieminen, J., Dhara, S., Chiu, W.C., Mucciolo, E.R., Bansil, A., 2023. Atomistic modeling of a superconductor–transition metal dichalcogenide–superconductor josephson junction. *Phys. Rev. B* 107, 174524. URL: <https://link.aps.org/doi/10.1103/PhysRevB.107.174524>, doi:10.1103/PhysRevB.107.174524.
- [102] Park, D., Kim, H., Kim, J., Kim, T., Lee, J., 2022. Snqs: scaling quantum circuit simulation using storage devices, in: *Proceedings of the 36th ACM International Conference on Supercomputing*, Association for Computing Machinery, New York, NY, USA. URL: <https://doi.org/10.1145/3524059.3532375>, doi:10.1145/3524059.3532375.
- [103] Plesch, M., Brukner, Č., 2011. Quantum-state preparation with universal gate decompositions. *Physical Review A—Atomic, Molecular, and Optical Physics* 83, 032302.
- [104] Popovici, D.T., Lee, H., Ben, M.D., Yoshioka, N., Ito, N., Klymko, K., Camps, D., Butko, A., 2025. A closeness centrality-based circuit partitioner for quantum simulations. URL: <https://arxiv.org/abs/2509.14098>, arXiv:2509.14098.
- [105] Putterman, H., Noh, K., Hann, C.T., MacCabe, G.S., Aghaieimodi, S., Patel, R.N., Lee, M., Jones, W.M., Moradinejad, H., Rodriguez, R., Mahuli, N., Rose, J., Owens, J.C., Levine, H., Rosenfeld, E., Reinhold, P., Monceli, L., Alcid, J.A., Alidoust, N., Arrangoiz-Arriola, P., Barnett, J., Bienias, P., Carson, H.A., Chen, C., Chen, L., Chinkezi, H., Chisholm, E.M., Chou, M.H., Clerk, A., Clifford, A., Cosmic, R., Curiel, A.V., Davis, E., DeLorenzo, L., D'Ewart, J.M., Diky, A., D'Souza, N., Dumitrescu, P.T., Eisenmann, S., Elkhoully, E., Evenbly, G., Fang, M.T., Fang, Y., Fling, M.J., Fon, W., Garcia, G., Gorshkov, A.V., Grant, J.A., Gray, M.J., Grimberg, S., Grimsom, A.L., Haim, A., Hand, J., He, Y., Hernandez, M., Hover, D., Hung, J.S.C., Hunt, M., Iverson, J., Jarrige, I., Jaskula, J.C., Jiang, L., Kalae, M., Karabalin, R., Karalekas, P.J., Keller, A.J., Khalajhedayati, A., Kubica, A., Lee, H., Leroux, C., Lieu, S., Ly, V., Madrigal, K.V., Marcaud, G., McCabe, G., Miles, C., Milsted, A., Minguzzi, J., Mishra, A., Mukherjee, B., Naghiloo, M., Obelpias, E., Ortuno, G., Pagdilao, J., Pancotti, N., Panduro, A., Paquette, J., Park, M., Peairs, G.A., Perello, D., Peterson, E.C., Ponte, S., Preskill, J., Qiao, J., Refael, G., Resnick, R., Retzker, A., Reyna, O.A., Runyan, M., Ryan, C.A., Sahnoud, A., Sanchez, E., Sanil, R., Sankar, K., Sato, Y., Scaffidi, T., Siavoshi, S., Sivarajah, P., Skogland, T., Su, C.J., Swenson, L.J., Teo, S.M., Tomada, A., Torlai, G., Wollack, E.A., Ye, Y., Zerrudo, J.A., Zhang, K., Brandão, F.G.S.L., Matheny, M.H., Painter, O., 2025. Hardware-efficient quantum error correction via concatenated bosonic qubits. *Nature* 638, 927–934. URL: <http://dx.doi.org/10.1038/s41586-025-08642-7>, doi:10.1038/s41586-025-08642-7.
- [106] Qing, M., Xie, W., 2023. Use vqe to calculate the ground energy of hydrogen molecules on ibm quantum. URL: <https://arxiv.org/abs/2305.06538>, arXiv:2305.06538.
- [107] Rakyta, P., Morse, G., Nádori, J., Majnay-Takács, Z., Mencer, O., Zimborás, Z., 2024. Highly optimized quantum circuits synthesized via data-flow engines. *Journal of Computational Physics* 500, 112756.
- [108] Ramkarthik, M., Solanki, P.D., 2021. Numerical recipes in quantum information theory and quantum computing: an adventure in FORTRAN 90. CRC Press.
- [109] Roffe, J., 2019. Quantum error correction: an introductory guide. *Contemporary Physics* 60, 226–245.
- [110] Rosenthal, G., 2023. Query and depth upper bounds for quantum unitaries via grover search. URL: <https://arxiv.org/abs/2111.07992>, arXiv:2111.07992.
- [111] Ryu, J., Jung, J., Lee, J., Gwon, Y., Kevin Rhee, J., 2021. A variational quantum algorithm for ordered svd, in: *2021 IEEE Global Communications Conference (GLOBECOM)*, pp. 01–05. doi:10.1109/GLOBECOM46510.2021.9685404.

- [112] Sawaya, N.P., Marti-Dafeik, D., Ho, Y., Tabor, D.P., Bernal Neira, D.E., Magann, A.B., Premaratne, S., Dubey, P., Matsuura, A., Bishop, N., De Jong, W.A., Benjamin, S., Parekh, O.D., Tubman, N.M., Klymko, K., Camps, D., 2023. Hamlib: A library of hamiltonians for benchmarking quantum algorithms and hardware, in: 2023 IEEE International Conference on Quantum Computing and Engineering (QCE), pp. 389–390. doi:[10.1109/QCE57702.2023.10296](https://doi.org/10.1109/QCE57702.2023.10296).
- [113] Sawaya, N.P., Smelyanskiy, M., McClean, J.R., Aspuru-Guzik, A., 2016. Error sensitivity to environmental noise in quantum circuits for chemical state preparation. *Journal of chemical theory and computation* 12, 3097–3108.
- [114] Seki, K., Kikuchi, Y., Hayata, T., Yunoki, S., 2025. Simulating floquet scrambling circuits on trapped-ion quantum computers. *Phys. Rev. Res.* 7, 023032. URL: <https://link.aps.org/doi/10.1103/PhysRevResearch.7.023032>, doi:[10.1103/PhysRevResearch.7.023032](https://doi.org/10.1103/PhysRevResearch.7.023032).
- [115] Shalf, J., 2020. The future of computing beyond moore’s law. *Philosophical Transactions of the Royal Society A* 378, 20190061.
- [116] Shende, V., Bullock, S., Markov, I., 2006. Synthesis of quantum-logic circuits. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 25, 1000–1010. doi:[10.1109/TCAD.2005.855930](https://doi.org/10.1109/TCAD.2005.855930).
- [117] Shinde, T., Budinski, L., Niemimäki, O., Lahtinen, V., Liebelt, H., Li, R., 2025. Utilizing classical programming principles in the intel quantum sdk: implementation of quantum lattice boltzmann method. *ACM Transactions on Quantum Computing* 6, 1–18.
- [118] Shirakawa, T., Ueda, H., Yunoki, S., 2024. Automatic quantum circuit encoding of a given arbitrary quantum state. *Phys. Rev. Res.* 6, 043008. URL: <https://link.aps.org/doi/10.1103/PhysRevResearch.6.043008>, doi:[10.1103/PhysRevResearch.6.043008](https://doi.org/10.1103/PhysRevResearch.6.043008).
- [119] Shor, P.W., 1995. Scheme for reducing decoherence in quantum computer memory. *Physical review A* 52, R2493.
- [120] Shor, P.W., 1999. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Review* 41, 303–332. URL: <https://doi.org/10.1137/S0036144598347011>, doi:[10.1137/S0036144598347011](https://doi.org/10.1137/S0036144598347011), arXiv:<https://arxiv.org/abs/10.1137/S0036144598347011>.
- [121] Srinivasan, S., Govindarajan, C.P., 2025. Exploring quantum measurements, observables and operators: Practical insights with amazon braket. URL: <https://aws.amazon.com/blogs/quantum-computing/exploring-quantum-measurements-observables-and-operators-practical-insights-with-amazon-braket/>.
- [122] Stetina, T., Wiebe, N., 2025. First-quantized quantum simulation of non-relativistic qed with emergent topologically protected coulomb interactions. URL: <https://arxiv.org/abs/2508.19343>, arXiv:2508.19343.
- [123] Sun, H., Galperin, M., 2025. Control of open quantum systems: The nonequilibrium green’s function perspective. *APL Quantum* 2.
- [124] Sun, X., Tian, G., Yang, S., Yuan, P., Zhang, S., 2023. Asymptotically optimal circuit depth for quantum state preparation and general unitary synthesis. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 42, 3301–3314.
- [125] team, Q.A., collaborators, 2020. qsim. URL: <https://doi.org/10.5281/zenodo.4023103>, doi:[10.5281/zenodo.4023103](https://doi.org/10.5281/zenodo.4023103).
- [126] Tilly, J., Chen, H., Cao, S., Picozzi, D., Setia, K., Li, Y., Grant, E., Wossnig, L., Rungger, I., Booth, G.H., et al., 2022. The variational quantum eigensolver: a review of methods and best practices. *Physics Reports* 986, 1–128.
- [127] Viamontes, G.F., Markov, I.L., Hayes, J.P., 2004. High-performance quidd-based simulation of quantum circuits, in: *Proceedings Design, Automation and Test in Europe Conference and Exhibition*, IEEE. pp. 1354–1355.
- [128] Wecker, D., Svore, K.M., 2014. LIQUi>: A Software Design Architecture and Domain-Specific Language for Quantum Computing. URL: [arXiv:1402.4467v1](https://arxiv.org/abs/1402.4467v1), arXiv:1402.4467.
- [129] Wille, R., Fowler, A., Naveh, Y., 2018. Computer-aided design for quantum computation, in: 2018 IEEE/ACM International Conference on Computer-Aided Design (ICCAD), IEEE. pp. 1–6.
- [130] Williams-Young, D.B., Asadchev, A., Popovici, D.T., Clark, D., Waldrop, J., Windus, T.L., Valeev, E.F., de Jong, W.A., 2023. Distributed memory, gpu accelerated fock construction for hybrid, gaussian basis density functional theory. *The Journal of Chemical Physics* 158.
- [131] Wintersperger, K., Dommert, F., Ehmer, T., Hoursanov, A., Klepsch, J., Mauerer, W., Reuber, G., Strohm, T., Yin, M., Luber, S., 2023. Neutral atom quantum computing hardware: performance and end-user perspective. *EPJ Quantum Technology* 10, 32.
- [132] Xu, M., Cao, S., Miao, X., Acar, U.A., Jia, Z., 2024. Atlas: Hierarchical partitioning for quantum circuit simulation on gpus, in: *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*, IEEE. pp. 1–17. doi:<https://doi.org/10.1109/SC41406.2024.00087>.
- [133] Zhang, C., Song, Z., Wang, H., Rong, K., Zhai, J., 2021. Hyquas: hybrid partitioner based quantum circuit simulation system on gpu, in: *Proceedings of the 35th ACM International Conference on Supercomputing*, pp. 443–454. doi:<https://doi.org/10.1145/3447818.3460357>.
- [134] Zhang, Q., Saligane, M., Kim, H.S., Blaauw, D., Tzimpragos, G., Sylvester, D., 2024. Quantum circuit simulation with fast tensor decision diagram, in: 2024 25th International Symposium on Quality Electronic Design (ISQED), IEEE. pp. 1–8.
- [135] Zhang, X.M., Li, T., Yuan, X., 2022. Quantum state preparation with optimal circuit depth: Implementations and applications. *Physical Review Letters* 129, 230504.
- [136] Zheng, W., Ying, L.A., Ding, P., 2004. Numerical solutions of the schrödinger equation for the ground lithium by the finite element method. *Applied mathematics and computation* 153, 685–695.
- [137] Žnidarič, M., Giraud, O., Georgeot, B., 2008. Optimal number of controlled-not gates to generate a three-qubit state. *Physical Review A—Atomic, Molecular, and Optical Physics* 77, 032320.
- [138] Zou, X., Lo, H.K., 2025. Algebraic topology principles behind topological quantum error correction. arXiv preprint arXiv:2505.06082 .
- [139] Zoufal, C., Lucchi, A., Woerner, S., 2019. Quantum generative adversarial networks for learning and loading random distributions. *npj Quantum Information* 5, 103.
- [140] Zylberman, J., Debbasch, F., 2024. Efficient quantum state preparation with walsh series. *Phys. Rev. A* 109, 042401. URL: <https://link.aps.org/doi/10.1103/PhysRevA.109.042401>, doi:[10.1103/PhysRevA.109.042401](https://doi.org/10.1103/PhysRevA.109.042401).